

LUCIDPREP

Grade 10 Biology

U.S. Standards-Aligned Course Syllabus

BIOLOGY

Biology · NGSS

36 weeks • 4 modules • 70% mastery

Curriculum Architecture

This course is organized as 36 weeks across four 9-week modules. Every week follows LucidPrep's established secondary learning cycle: AI Instructor opening, objectives, vocabulary, five warm-ups, main lecture, two short tutorial videos, AI Instructor presentation, two homework sets, and a 40-question assessment.

Mastery standard: 70%+. Students below mastery review the instructional presentation and may retake the assessment, matching the existing LucidPrep secondary model.

U.S. Alignment Note

There is no single national science curriculum, but the NGSS are the dominant U.S. reference. Biology is the most common Grade 10 science course. Because high-school science sequencing varies (Biology may sit in Grade 9 or 10), this Biology course mirrors the Biology option presented in the Grade 9 curriculum; a school places it once, in whichever grade its sequence uses.

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Common Instructional & Assessment Model

Weekly Learning Cycle

Step	Component	Description
1	Opening / Domain Overview	~2-minute AI Instructor introduction
2	Objectives + Vocabulary	Student-facing outcomes and core terminology
3	Warm-Up	5 questions with explanations
4	Main Lecture	Concept instruction, models and examples
5	Video Tutorials	2 short instructional videos
6	AI Instructor Presentation	Visual models, examples, strategies and applications
7	Homework	Two sets of 5 questions
8	Weekly Assessment	40 questions; 70% mastery
9	Review / Retake	Below mastery → review presentation → retake

Assessment Philosophy

Assessment is continuous and mastery-based. Each week pairs low-stakes practice (five warm-ups and two homework sets) with a 40-question weekly assessment. Students scoring below 70% review the instructional presentation and retake, so mastery is reached before progressing. Module assessments at Weeks 9, 18 and 27 and a cumulative final at Week 36 measure retention and synthesis.

This 36-week course is designed around **the NGSS High School Life Science standards (HS-LS1-HS-LS4), with Science & Engineering Practices and Crosscutting Concepts**. It is a full-year curriculum with four nine-week modules and a consistent LucidPrep mastery workflow.

Grade	Grade 10
Course	Biology · NGSS
Duration	36 weeks
Structure	4 modules × 9 weeks
Weekly assessment	40 questions
Mastery threshold	70%
Module assessments	Weeks 9, 18, 27
Final assessment	Week 36

Course Learning Outcomes

- Explain systems from molecules and cells through populations and ecosystems.
- Use models, data and evidence to explain biological phenomena.
- Analyze inheritance, gene expression, evolution and ecological interactions.
- Conduct and evaluate scientific investigations.
- Communicate scientific explanations with accurate terminology and reasoning.

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Module 1 — Foundations of Biology: Chemistry, Cells & Organization

Week	Topics	Students will be able to	Standards
Week 1 — Biology & Scientific Investigation	Characteristics of life; observations vs. inferences; hypotheses; variables; controls; experimental design; models	Identify variables and controls; construct evidence-based conclusions	SEP: questions, investigations, data; CCC: cause/effect
Week 2 — Chemistry of Life	Matter; elements; atoms; isotopes; ions; ionic/covalent bonds; polarity	Describe atomic structure; relate valence electrons to behavior	HS-PS1 foundation
Week 3 — Water & Biological Molecules	Water polarity; hydrogen bonding; pH; buffers; carbohydrates; lipids; proteins; nucleic acids	Classify macromolecules; connect structure to function	Structure/function; matter and energy
Week 4 — Cell Structure & Function	Cell theory; prokaryotic/eukaryotic; plant/animal cells; organelles; specialization	Compare cell types; explain organelle functions	HS-LS1 structure/function
Week 5 — Membranes & Transport	Selective permeability; diffusion; osmosis; facilitated diffusion; active transport; endo/exocytosis	Predict membrane movement; explain homeostasis	HS-LS1 interacting systems
Week 6 — Energy & Enzymes	ATP; activation energy; enzymes; active sites; temperature; pH; concentration	Analyze enzyme data; connect ATP to cellular work	HS-LS1 energy/matter
Week 7 — Cellular Respiration	Glycolysis; Krebs cycle; electron transport; aerobic/anaerobic respiration; fermentation	Trace energy transfer; compare pathways	HS-LS1 energy/matter
Week 8 — Photosynthesis	Chloroplasts; pigments; light reactions; Calvin cycle; carbon fixation	Model photosynthesis; connect to respiration	HS-LS1/LS2 energy
Week 9 — Module Assessment	Integrated review of chemistry, cells, transport, enzymes, respiration and photosynthesis	Synthesize concepts; demonstrate 70%+ mastery	Performance assessment

Module delivery notes. Use diagrams and models heavily. Favor cause-and-effect reasoning, biological data interpretation and real-world phenomena such as disease, resistance, biodiversity and ecosystem change.

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Module 2 — Genetics, DNA & Cellular Reproduction

Week	Topics	Students will be able to	Standards
Week 1 — Cell Cycle & Mitosis	Cell cycle; interphase; chromosomes; mitosis; cytokinesis; cancer	Model division; explain growth; connect uncontrolled division to cancer	HS-LS1 cell division
Week 2 — Meiosis & Sexual Reproduction	Meiosis; haploid/diploid; gametes; homologous chromosomes; crossing over; assortment	Compare mitosis/meiosis; explain variation	HS-LS3 inheritance
Week 3 — DNA Structure & Replication	Nucleotides; double helix; base pairing; genes; chromosomes; replication	Model DNA; explain replication	HS-LS3 information
Week 4 — RNA & Transcription	mRNA; tRNA; rRNA; transcription; codons; RNA processing	Explain transcription; interpret codons	HS-LS1/LS3 information flow
Week 5 — Protein Synthesis	Translation; ribosomes; amino acids; DNA → RNA → protein	Model translation; connect DNA to proteins	HS-LS1-1
Week 6 — Mendelian Genetics	Alleles; genotype/phenotype; dominant/recessive; Punnett squares; probability	Solve monohybrid crosses; predict inheritance	HS-LS3
Week 7 — Beyond Mendel	Incomplete dominance; codominance; multiple alleles; sex-linked; polygenic traits	Analyze non-Mendelian patterns	State biology framework extension
Week 8 — Mutations & Biotechnology	Substitution; insertion/deletion; frameshift; disorders; genetic engineering; ethics	Classify mutations; predict effects; evaluate biotech scenarios	HS-LS3 variation
Week 9 — Module Assessment	Cell division, DNA/RNA, proteins, inheritance, variation	Integrate genetics; demonstrate 70%+ mastery	Performance assessment

Module delivery notes. Use diagrams and models heavily. Favor cause-and-effect reasoning, biological data interpretation and real-world phenomena such as disease, resistance, biodiversity and ecosystem change.

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Module 3 — Evolution, Natural Selection & Biological Diversity

Week	Topics	Students will be able to	Standards
Week 1 — Evidence for Evolution	Fossils; anatomy; molecular evidence; embryology; biogeography	Evaluate lines of evidence; infer ancestry	HS-LS4 evidence
Week 2 — Natural Selection	Variation; heritability; competition; selection pressure; differential reproduction	Explain population change and adaptation	HS-LS4 natural selection
Week 3 — Evolutionary Mechanisms	Mutation; gene flow; drift; founder/bottleneck; recombination; allele frequencies	Compare mechanisms; interpret population data	HS-LS4 mechanisms
Week 4 — Speciation	Species; geographic/behavioral/temporal isolation; divergence; adaptive radiation	Model speciation; identify barriers	HS-LS4 species formation
Week 5 — Classification & Phylogeny	Taxonomy; domains; cladograms; phylogenetic trees; derived traits	Read phylogenies; infer relatedness	Modeling/evidence
Week 6 — Biodiversity	Genetic/species/ecosystem diversity; extinction; conservation	Analyze threats; connect diversity and stability	HS-LS2/LS4
Week 7 — Adaptation & Coevolution	Structural/behavioral/physiological adaptations; mimicry; predator-prey; host-parasite	Analyze adaptations and coevolution	HS-LS4
Week 8 — Evolution in Action	Antibiotic resistance; pesticide resistance; viral evolution; artificial selection	Analyze real-world evolution data	HS-LS4 applications
Week 9 — Module Assessment	Evolutionary evidence, selection, mechanisms, speciation, biodiversity	Synthesize; demonstrate 70%+ mastery	Performance assessment

Module delivery notes. Use diagrams and models heavily. Favor cause-and-effect reasoning, biological data interpretation and real-world phenomena such as disease, resistance, biodiversity and ecosystem change.

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Module 4 — Ecology, Ecosystems & Sustainability

Week	Topics	Students will be able to	Standards
Week 1 — Ecological Organization	Organism; population; community; ecosystem; biome; biosphere; habitat; niche	Identify levels; distinguish habitat/niche	HS-LS2
Week 2 — Energy Flow	Producers; consumers; decomposers; food webs; trophic levels; energy pyramids	Trace energy; interpret food webs	HS-LS2 energy transfer
Week 3 — Matter Cycling	Water; carbon; nitrogen; phosphorus; decomposition; human impacts	Model cycles; explain conservation of matter	HS-LS2 matter cycling
Week 4 — Population Ecology	Density; distribution; birth/death; migration; exponential/logistic growth; carrying capacity	Model population growth; interpret data	HS-LS2 population dynamics
Week 5 — Community Ecology	Competition; predation; herbivory; parasitism; mutualism; commensalism; niche	Classify interactions; predict outcomes	HS-LS2 interactions
Week 6 — Ecosystem Dynamics	Disturbance; succession; resilience; limiting factors; feedback	Model ecosystem change; evaluate resilience	HS-LS2 stability/change
Week 7 — Human Impact	Habitat loss; pollution; climate change; invasive species; overharvesting	Analyze environmental impacts and evidence	HS-LS2/ESS3
Week 8 — Sustainability & Conservation	Renewable resources; agriculture; water; energy; restoration; conservation	Evaluate solutions; propose evidence-based actions	Engineering/Earth-s system connections
Week 9 — Final Assessment	Full-course synthesis: molecules → cells → organisms → populations → ecosystems	Solve unfamiliar scenarios; demonstrate mastery	Final performance assessment

Module delivery notes. Use diagrams and models heavily. Favor cause-and-effect reasoning, biological data interpretation and real-world phenomena such as disease, resistance, biodiversity and ecosystem change.

Assessment & Alignment

Assessment Blueprint

Warm-ups: weekly • Homework: weekly • Weekly assessment: 40 questions • Module assessments: Weeks 9 / 18 / 27 • Final: Week 36 • Below 70%: instructional review and retake.

Standards Alignment

Primary framework	NGSS: HS-LS1, HS-LS2, HS-LS3, HS-LS4
Science & Engineering Practices	Asking questions and defining problems; developing and using models; planning and carrying out investigations; analyzing and interpreting data; using mathematics and computational thinking; constructing explanations and designing solutions; engaging in argument from evidence; obtaining, evaluating and communicating information.
Crosscutting Concepts	Patterns; cause and effect; scale, proportion and quantity; systems and system models; energy and matter; structure and function; stability and change.
State mapping	NGSS adopted or adapted by nearly all states; middle-school standards are banded 6–8 and arranged differently by state.

Content Production Guidance

Author each week around an anchoring phenomenon or problem: 5 warm-ups, an explanatory lecture with models, 2 videos, an AI presentation with visual models and investigations, 2 homework sets, and a 40-question bank mixing three-dimensional items (SEP × DCI × CCC). Tag items to NGSS performance expectations.

Curriculum Basis & Source Notes

This Grade 10 science syllabus was generated from the Next Generation Science Standards. It reflects Biology, the most common Grade 10 science course; the same content appears as the Biology option in the Grade 9 science curriculum, since HS science placement varies by district.

Primary reference	Next Generation Science Standards — HS Life Science performance expectations (HS-LS1–LS4), with Science & Engineering Practices and Crosscutting Concepts.
Framework basis	A Framework for K-12 Science Education (National Research Council).
State cross-check	California and other NGSS-adopting state science frameworks.
LucidPrep alignment	Mapped to LucidPrep's 36-week, 4-module structure with 70% mastery.

Recommended next production step. Select anchoring phenomena per module and commission three-dimensional item banks (SEP × DCI × CCC) tagged to NGSS performance expectations.